ANSWER KEY

English Test

1. C	21. D	41. A	61. B
2. F	22. H	42. J	62. J
3. B	23. D	43. C	63. A
4. H	24. F	44. H	64. J
5. D	25. B	45. A	65. A
6. H	26. F	46. J	66. G
7. D	27. D	47. B	67. B
8. F	28. H	48. G	68. J
9. B	29. A	49. D	69. C
10. G	30. G	50. F	70. H
11. D	31. D	51. B	71. B
12. H	32. G	52. H	72. F
13. C	33. A	53. B	73. B
14. F	34. H	54. H	74. J
15. B	35. A	55. C	75. C
16. G	36. J	56. J	
17. D	37. B	57. A	
18. F	38. H	58. F	
19. B	39. B	59. B	
20. H	40. J	60. H	

Mathematics Test

1. C	21. D	41. D
2. F	22. H	42. H
3. B	23. C	43. A
4. J	24. G	44. G
5. D	25. C	45. D
6. H	26. G	46. F
7. D	27. E	47. C
8. K	28. J	48. F
9. A	29. B	49. C
10. J	30. H	50. K
11. C	31. B	51. B
12. G	32. H	52. F
13. C	33. D	53. E
14. J	34. J	54. K
15. A	35. E	55. C
16. H	36. F	56. J
17. E	37. D	57. E
18. G	38. G	58. J
19. A	39. E	59. D
20. J	40. H	60. K

Reading Test

1. C	21. B
2. G	22. H
3. D	23. C
4. G	24. H
5. C	25. A
6. H	26. J
7. B	27. D
8. J	28. G
9. A	29. C
10. J	30. G
11. A	31. C
12. G	32. G
13. B	33. A
14. J	34. H
15. C	35. D
16. J	36. J
17. A	37. A
18. H	38. H
19. C	39. B
20. F	40. G

Science Reasoning Test

1. A	21. C
2. H	22. G
3. A	23. A
4. J	24. F
5. B	25. C
6. J	26. J
7. D	27. A
8. H	28. F
9. C	29. C
10. J	30. H
11. B	31. A
12. G	32. J
13. A	33. D
14. G	34. H
15. C	35. B
16. H	36. G
17. B	37. C
18. G	38. H
19. D	39. B
20. H	40. H

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SCORING GUIDE

Your final reported score is your COMPOSITE SCORE. Your COMPOSITE SCORE is the average of all of your SCALE SCORES.

Your SCALE SCORES for the four multiple-choice sections are derived from the Scoring Table on the next page. Use your RAW SCORE, or the number of questions that you answered correctly for each section, to determine your SCALE SCORE. If you got a RAW SCORE of 60 on the English test, for example, you correctly answered 60 out of 75 questions.

Step 1 Determine your RAW SCORE for each of the four multiple-choice sections:

English _____

Mathematics _____

Reading _____

Science Reasoning _____

The following Raw Score Table shows the total possible points for each section.

RAW SCORE TABLE	
KNOWLEDGE AND SKILL AREAS	**RAW SCORES**
ENGLISH	75
MATHEMATICS	60
READING	40
SCIENCE REASONING	40
WRITING	12

Multiple-Choice Scoring Worksheet

Step 2 Determine your SCALE SCORE for each of the four multiple-choice sections using the following Scoring Worksheet. Each SCALE SCORE should be rounded to the nearest number according to normal rules. For example, 31.2 ≈ 31 and 31.5 ≈ 32. If you answered 61 questions correctly on the English section, for example, your SCALE SCORE would be 29.

English	_____ RAW SCORE	× **36** =	_____	÷ **75** =	_____	**− 2**	(*correction factor)
					_____ SCALE SCORE		
Mathematics	_____ RAW SCORE	× **36** =	_____	÷ **60** =	_____	**+ 1**	(*correction factor)
					_____ SCALE SCORE		
Reading	_____ RAW SCORE	× **36** =	_____	÷ **40** =	_____	**+ 2**	(*correction factor)
					_____ SCALE SCORE		
Science Reasoning	_____ RAW SCORE	× **36** =	_____	÷ **40** =	_____	**+ 1.5**	(*correction factor)
					_____ SCALE SCORE		

*The correction factor is an approximation based on the average from several recent ACT tests. It is most valid for scores in the middle 50% (approximately 16–24 scale composite score) of the scoring range.

The scores are all approximate. Actual ACT scoring scales vary from one administration to the next based upon several factors.

If you take the optional Writing Test, you will need to combine your English and Writing scores to obtain your final COMPOSITE SCORE. Once you have determined a score for your essay out of 12 possible points, you will need to determine your ENGLISH/WRITING SCALE SCORE, using both your ENGLISH SCALE SCORE and your WRITING TEST SCORE. The combination of the two scores will give you an ENGLISH/WRITING SCALE SCORE, from 1 to 36, that will be used to determine your COMPOSITE SCORE mentioned earlier.

Using the English/Writing Scoring Table, find your ENGLISH SCALE SCORE on the left or right hand side of the table and your WRITING TEST SCORE on the top of the table. Follow your ENGLISH SCALE SCORE over and your WRITING TEST SCORE down until the two columns meet at a number. This number is your ENGLISH/WRITING SCALE SCORE and will be used to determine your COMPOSITE SCORE.

Step 3 Determine your ENGLISH/WRITING SCALE SCORE using the English/Writing Scoring Table on the following page:

English _____

Writing _____

English/Writing _____

ENGLISH/WRITING SCORING TABLE

ENGLISH SCALE SCORE	WRITING TEST SCORE											ENGLISH SCALE SCORE
	2	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	
36	26	27	28	29	30	31	32	33	34	32	36	**36**
35	26	27	28	29	30	31	31	32	33	34	35	**35**
34	25	26	27	28	29	30	31	32	33	34	35	**34**
33	24	25	26	27	28	29	30	31	32	33	34	**33**
32	24	25	25	26	27	28	29	30	31	32	33	**32**
31	23	24	25	26	27	28	29	30	30	31	32	**31**
30	22	23	24	25	26	27	28	29	30	31	32	**30**
29	21	22	23	24	25	26	27	28	29	30	31	**29**
28	21	22	23	24	24	25	26	27	28	29	30	**28**
27	20	21	22	23	24	25	26	27	28	28	29	**27**
26	19	20	21	22	23	24	25	26	27	28	29	**26**
25	18	19	20	21	22	23	24	25	26	27	28	**25**
24	18	19	20	21	22	23	23	24	25	26	27	**24**
23	17	18	19	20	21	22	23	24	25	26	27	**23**
22	16	17	18	19	20	21	22	23	24	25	26	**22**
21	16	17	17	18	19	20	21	22	23	24	25	**21**
20	15	16	17	18	19	20	21	21	22	23	24	**20**
19	14	15	16	17	18	19	20	21	22	23	24	**19**
18	13	14	15	16	17	18	19	20	21	22	23	**18**
17	13	14	15	16	16	17	18	19	20	21	22	**17**
16	12	13	14	15	16	17	18	19	20	20	21	**16**
15	11	12	13	14	15	16	17	18	19	20	21	**15**
14	10	11	12	13	14	15	16	17	18	19	20	**14**
13	10	11	12	13	14	14	15	16	17	18	19	**13**
12	9	10	11	12	13	14	15	16	17	18	19	**12**
11	8	9	10	11	12	13	14	15	16	17	18	**11**
10	8	9	9	10	11	12	13	14	15	16	17	**10**
9	7	8	9	10	11	12	13	13	14	15	16	**9**
8	6	7	8	9	10	11	12	13	14	15	16	**8**
7	5	6	7	8	9	10	11	12	13	14	15	**7**
6	5	6	7	7	8	9	10	11	12	13	14	**6**
5	4	5	6	7	8	9	10	11	12	12	13	**5**
4	3	4	5	6	7	8	9	10	11	12	13	**4**
3	2	3	4	5	6	7	8	9	10	11	12	**3**
2	2	3	4	5	6	6	7	8	9	10	11	**2**
1	1	2	3	4	5	6	7	8	9	10	11	**1**

Step 4 Determine your COMPOSITE SCORE by finding the sum of all your SCALE SCORES for each of the four sections: English only (if you do not choose to take the optional Writing Test) *or* English/Writing (if you choose to take the optional Writing Test), Math, Reading, and Science Reasoning, and divide by 4 to find the average. Round your COMPOSITE SCORE according to normal rules. For example, 31.2 ≈ 31 and 31.5 ≈ 32.

_____	+	_____	+	_____	+	_____	=	_____
ENGLISH *OR* ENGLISH/WRITING SCALE SCORE		MATHEMATICS SCALE SCORE		READING SCALE SCORE		SCIENCE SCALE SCORE		SCALE SCORE TOTAL

_____	÷ 4 =	_____
SCALE SCORE TOTAL		COMPOSITE SCORE

ANSWERS AND EXPLANATIONS

English Test Explanations

PASSAGE I

1. **The best answer is C.** It is necessary to include a comma between "tedious" and "tiring" because they are coordinate adjectives—they modify the same noun. Answer choice A is incorrect because it includes two additional and unnecessary commas.

2. **The best answer is F.** The sentence is clear as it is written. The other answer choices would be very awkward to read.

3. **The best answer is B.** This question relies on your ability to recognize wordiness. The phrase "needless to say" adds little to the meaning to the sentence and interrupts the natural rhythm of the clause.

4. **The best answer is H.** This question relies on your ability to recognize redundancy. "Asking" is equivalent to "inquiring," so answer choices F and G may be eliminated. Between answer choices H and J, answer choice H is better because it omits the unnecessary phrase "of me," further eliminating wordiness.

5. **The best answer is D.** The future tense is necessary because the feeling of passion the hiker asserts will come only after reading the book.

6. **The best answer is H.** The verb "brings" requires a direct object (what is being brought) and a prepositional element (where the direct object is being brought). The pronoun "you" and the phrase "closer to peace" are the constituents of the verb form "bring"; no commas are necessary within the clause.

7. **The best answer is D.** The preceding sentence identifies what is "no longer" the case. It would make sense that the next sentence describes what is true currently. Answer choices B and D may appear to be equally appropriate, except answer choice B must be eliminated because "now" appears again later in the sentence.

8. **The best answer is F.** Placement before a noun indicates that the word must be an adjective, eliminating answer choices G and J.

9. **The best answer is B.** The word "time" should refer to "sunrises and sunsets," so the plural "times" is more appropriate; therefore, answer choices A and D may be eliminated. Answer choice C does not have correct word order, so it may be eliminated.

10. **The best answer is G.** Nouns conjoined with "or," as in the underlined portion, stand on equal footing. They should not have any intervening commas.

11. **The best answer is D.** The word "notepad" implies writing, which is the subject of the next paragraph. As a description of how moments of literary creativity arise for the author, this answer choice would provide a good transition between paragraphs.

12. **The best answer is H.** The phrase "other times" means "on other occasions." Among the answer choices, only H is acceptable because it is plural and lacks the possessive apostrophe.

13. **The best answer is C.** Answer choice A may be eliminated for wordiness. Answer choice B is an adjective, so it cannot modify a verb; eliminate it. Answer choices C and D would both be grammatical, but answer choice C would form a less awkward chain of modifiers.

14. **The best answer is F.** The word "others" is a noun meaning "other people;" eliminate answer choices H and J. As the passage in written in present tense, the past tense answer choice G may be eliminated.

15. **The best answer is B.** This sentence is simply a response to the preceding sentence, which identifies some people's preferred activity. The introductory phrase "For me" is a natural fit for a sentence about the author's opposing view.

PASSAGE II

16. **The best answer is G.** The word "ever" is best placed between auxiliary and participle; eliminate answer choice J. The word "saw" is the simple past form of "see," not the participle; therefore, answer choice H may be eliminated.

17. **The best answer is D.** The word "urban" is an adjective and should modify the noun "streets." Answer choices A and B contain an incorrect possessive form. Answer choice C may be eliminated because adjectives do not take the pluralizing "s" before nouns.

18. **The best answer is F.** The next-best answer choice would be J because both answer choices F and J have appropriate meaning and word order; however, answer choice J may be eliminated because the clause refers to vacation in general, so it must be written in simple present tense.

19. **The best answer is B.** The next two paragraphs explore the differences between dogs in America and dogs in Africa. Answer choice B provides a general introduction to what is coming up in the essay.

20. **The best answer is H.** The dog is panting, not the owner, so answer choices F and G may be eliminated. The sentence is in simple present tense, so answer choice H is best.

21. **The best answer is D.** "Dog owners" is a compound noun and the subject of the sentence. The other answer choices incorrectly contain possessive markers on one or both parts of the compound.

22. **The best answer is H.** The preceding sentence asserts people who have never owned a dog think dogs are not smart. Only answer choice H reveals naïveté on the part of the animal.

23. **The best answer is D.** To get a dog to improve behavior, it is logical that more training is needed, not less.

24. **The best answer is F.** The sentence addresses a general situation and makes sense in the simple present tense. Answer choice F is the third-person singular simple present form of "happen," and agrees with the subject "a dog."

25. **The best answer is B.** This clause should parallel the previous clause's logical subject ("a dog" becomes "it") and tense; therefore, the other answer choices may be eliminated.

26. **The best answer is F.** "Stark" is an adjective commonly used with "differences" in order to emphasize the degree of difference. Here, "starkly" (adv.) modifies "different" (adj.), so answer choice F is best. The other answer choices have awkward word order.

27. **The best answer is D.** "Mutualism" is a kind of symbiotic relationship that exists in nature. To say that "mutualism" between person and dog is tangible means that the exchange of benefits between the two is outwardly apparent. Inserting "tangible" before "mutualism" is supported by the rest of the paragraph, which details exactly how the human and dog profit from each other's presence.

28. **The best answer is H.** The "but" heads information that is part of the clause. Answer choice F may be eliminated because a comma is necessary before "but." Answer choice G may be eliminated because the semicolon divides only full independent clauses. Answer choice J may be eliminated because the dash is used to give special emphasis to something, which is not appropriate here.

29. **The best answer is A.** Answer choices B and C may be eliminated first because they are very awkward. Answer choice D, while less awkward, should be eliminated, too, because the placement of "never" after the verb is less acceptable than the placement of "never" at the beginning of the sentence, as in answer choice A.

30. **The best answer is G.** A single comma preceding "so" is all that is necessary. No comma is necessary after "so" when it introduces an action resulting from some pre-existing condition (e.g., "It rained, so I stayed home."); therefore, answer choices H and J may be eliminated.

PASSAGE III

31. **The best answer is D.** This sentence states something that seems at odds with the preceding sentence. Only "though," answer choice D, reflects this relationship.

32. **The best answer is G.** "Carnaval season" indicates that the event comes around at least once a year. The verb "flock" should be in simple present tense to reflect how the clause is a general statement.

33. **The best answer is A.** The sentence is clearest as written. Answer choices B and C are comma-spliced run-on sentences. The subject-predicate pair in answer choice D does not make sense.

34. **The best answer is H.** Reminding the reader of the dangers the author likes to face emphasizes his "adventurous nature," as the question puts it. The other answer choices do not seem well-matched to the "extreme" theme the author maintains throughout the passage.

35. **The best answer is A.** The major clue here is the initial "If" clause in the sentence. It must be followed by a comma, then a second clause; the other answer choices may be eliminated.

36. **The best answer is J.** Answer choice F has correct word order, but a comma would be necessary before the gerund "shooting." The best alternative

is answer choice J because it uses the conjunction "and," which does not require a preceding comma, and maintains past tense from the preceding clause.

37. The best answer is B. The "sarcastic grin" from the preceding sentence surely indicates that the race organizer has his doubts about the author's fitness for the activity, but is willing to let him try it nonetheless. Only answer choice B reflects this attitude.

38. The best answer is H. The verb "went" may only be modified by an adverb, so the other answer choices should be eliminated.

39. The best answer is B. Answer choice C may be eliminated for being a sentence fragment. The other answer choices are grammatical, but answer choice B is clearest because it is in conventional word order and has active voice.

40. The best answer is J. The phrase "This is the time" is wordy; eliminate answer choice F. The preceding sentence does a satisfactory job of defining the time frame. No words are required before the main clause "I learned … legal"; therefore, answer choices G and H may be eliminated.

41. The best answer is A. Here, "Racing" is the subject of the sentence, so answer choice C may be eliminated. Since the race took place in the past, answer choice A is appropriate.

42. The best answer is J. Because that the race result was so close, the mention of a photo finish seems an appropriate place for an exclamation point. Answer choice F leaves a sentence fragment at the end of the paragraph. Answer choices G and H may be eliminated because they incorrectly link the two clauses, which are not related conceptually.

43. The best answer is C. The noun phrase "head of the pack" is a common colloquialism and should not be rephrased.

44. The best answer is H. This answer choice ties the events of the author's heat with the overarching concept of his penchant for thrill seeking. Surely, the author would celebrate a successful slide down the chute.

45. The best answer is A. An "essay discussing extreme sports" would have to include information on several extreme sports. This essay only mentions one.

PASSAGE IV

46. The best answer is J. What follows a semicolon dividing a sentence must be a well-formed clause, containing minimally a subject and a verb. Answer choice J correctly uses "the water" as a subject and "shines" as the appropriate simple present tense verb. Answer choice G would make a well-formed clause, but the past tense is not appropriate for giving a description of a place that is unlikely to be affected by time (like the color of the water).

47. The best answer is B. The possessive form of the pronoun "it" is "its." Note the lack of apostrophe, which contradicts the normal rule of adding apostrophe + "s" to singular nouns to create possessive forms.

48. The best answer is G. The sentence means to say that the "euphoric swell" is responsible for pushing up "the corners of the mouth" of some hypothetical visitor to the Caribbean. Answer choice G uses conventional word order and is clearest and most concise.

49. The best answer is D. The verb phrase from the previous sentence, "sailed blissfully from one cay to another," means exactly what this sentence does. For the sake of brevity, it may be omitted.

50. The best answer is F. This paragraph describes the chaotic scene of a sailing crew effecting a change of direction. Answer choice F alludes to this fact by saying that tacking is "reserved for the experts."

51. The best answer is B. A verb in infinitive or gerund form would be appropriate here. Answer choice A lacks "to" before the verb in order to form the infinitive. Answer choice C may be eliminated because "start" is redundant with "begins." Answer choice D is not appropriate because omitting the mention of yelling takes away from the chaotic scene the author is trying to create.

52. The best answer is H. For the sake of clarity and brevity, the simple infinitive form should be used here.

53. The best answer is B. A clause introduced by a subordinating conjunctive adverb such as "along with" must be set apart from the sentence with commas. Eliminate answer choices A and D. Answer choice C is incorrect because the past tense verb "went" is not appropriate.

54. The best answer is H. This answer choice uses conventional word order and is clearest and most concise.

55. The best answer is C. Introducing speech or inner monologue requires a comma before the direct quotation. Furthermore, if the quotation is a complete sentence, the first word of it should be capitalized as usual.

56. The best answer is J. The subject "the unpredictable wind" introduces the main clause, so what comes before it should anticipate that. Answer choices F and H fail to do this, and answer choice G is ungrammatical; they all may be eliminated.

57. The best answer is A. This sentence is part of the narrative of the author's trip, so it should be in past tense; eliminate the other answer choices.

58. The best answer is F. The preceding two sentences identify that the boat lay adrift at sea for some time before the motor was started. The introductory word "Eventually," answer choice F, is most appropriate.

59. The best answer is B. This sentence is part of the narrative of the author's trip, so it should be in past tense; eliminate answer choices A and C. Answer choice D may be eliminated because it does not make sense in context.

60. The best answer is H. The sentence gives the author's ultimate reaction to the absence of wind at the end of his sailing trip. It provides a conclusion to the details of the paragraph.

PASSAGE V

61. The best answer is B. In general, details introduced by the word "including" must be set apart from the clause with a comma.

62. The best answer is J. A subordinate clause headed by "but" or "and" must be set apart from the main clause with a comma. Eliminate answer choice H. Answer choice G would create a run-on sentence; eliminate it. Answer choice J is the best fit for the sentence because the adverb "ordinarily" describes a general condition that the clause with "but" is going to counter. Careful reading will reveal that "but" (answer choice J) is preferable to "and" (answer choice F) in the sentence.

63. The best answer is A. These two sentences are independent thoughts and cannot be separated with a colon or a comma; eliminate answer choices B and D. The adverb "thus" normally introduces the consequence of some pre-stated condition. The two sentences addresses by the questions do not have that relationship.

64. The best answer is J. If a clause comes after a form of the verb "decide," either the complementizer "that" or a null (absent) complementizer must precede the clause.

65. The best answer is A. Answer choice B may be eliminated for wordiness. Answer choice C may be eliminated for the lack of a conjunction between any of the elements of the list, as well as incorrect use of the comma after "Titian." The "*Mona Lisa*" is a painting and "Titian" and "Raphael" are artists, so answer choice D does not make sense.

66. The best answer is G. At the beginning of the paragraph, the Louvre is described as "majestic." Certainly, a the French king would want a palace to match his extensive art collection. The other answer choices do not reflect this, so they may be eliminated.

67. The best answer is B. Charles V is mentioned earlier in the passage as a previous resident of the Louvre. Answer choice B describes a durative condition—how nice the palace was—in the past.

68. The best answer is J. It is generally inappropriate to begin a paragraph with a sentence that uses one or more pronouns; therefore, answer choice J would create the clearest sentence.

69. The best answer is C. This sentence pairs the number of works of art with the corresponding year and represents the clearest word order among all the choices.

70. The best answer is H. The concept of "elitism" matches best with "creativity" because they are both human qualities. The other answer choices are words for certain kinds of people and would not be a good fit in the conjoined phrase.

71. The best answer is B. The preceding paragraph focuses on Francois 1 and his renovation of the Louvre. Therefore, Sentence 1 is a good transition between the third and fourth paragraphs. Eliminate answer choice D, which places Sentence 2 first. The three remaining answer choices (A, B, and C) all put Sentence 2 in the second position, so look at whether Sentence 3, Sentence 4, or Sentence 5 should follow Sentence 2. Because Sentence 2 mentions that royalty lived in the sprawling palace (the Louvre), Sentence 3, which discusses the art collection, seems out of place. Eliminate answer choice A. Sentence 5 effectively joins Sentence 2 and 6 by indicating that after Louis XIV left, many more people lived in the palace. Sentences 3 and

4 chronologically follow Sentence 6, which makes answer choice B the most logical selection.

72. The best answer is F. The "ballooning collection" mentioned in this paragraph supports inserting an introductory sentence like answer choice F at the head of the paragraph. None of the other answer choices are supported by evidence in the paragraph.

73. The best answer is B. The correct idiomatic phrase is "moved to," which indicates where the artwork has been moved.

74. The best answer is J. Answer choice J creates two incomplete sentences, so it would not be acceptable. The remaining answer choices are grammatically correct, and do not change the meaning of the paragraph.

75. The best answer is C. The verb should match the unconjugated form of "visit" before the conjunction "and;" eliminate answer choice A. Answer choice B shows "view" as a noun, so it may be eliminated. Answer choice D may be eliminated since it would leave "and" without a second part.

Mathematics Test Explanations

1. **The correct answer is C.** Since the point (2,−5) has a positive x-coordinate, it will have a placement to the right of the y-axis. Similarly, since the point has a negative y-coordinate, it will have a placement below the x-axis. Of the possible points, only C and D fit such a description. Of the two points, C appears to have a y-coordinate with greater magnitude than its x-coordinate, which would correspond to the point (2,−5).

2. **The correct answer is F.** To solve this problem, systematically assess each answer choice and eliminate incorrect choices. Of the choices, only the statement that "Michael has a student ID" must be true because it is given that "Michael is a high school student who attends Tarrytown High School" and "All students who attend Tarrytown High School have a student ID."

3. **The correct answer is B.** When Juan withdrew \$300, the resulting balance was \$400, making the balance of the account before withdrawal \$400 + \$300, or \$700. Since the balance of Juan's savings account quadrupled during the year, the balance, b, in the account before it quadrupled is represented by $4b = 700$. Therefore $b = \frac{700}{4}$, or \$175.

4. **The correct answer is J.** To find the value of a such that the equation $3(a + 5) - a = 23$ is true, solve $3(a + 5) - a = 23$ for a:

$$3a + 15 - a = 23$$
$$2a + 15 = 23$$
$$2a = 8$$
$$a = 4$$

5. **The correct answer is D.** It is given that on the number line, numbers decrease in value from right to left, meaning that numbers on the right are greater than numbers on the left. Since no indication of the location of 0 is given in the picture, no assumption about negative or positive can be made. Simply because numbers on the right are greater than numbers on the left and Y is to the right of X, Y must be greater than X. Put another way, X must be less than Y, answer choice D.

6. **The correct answer is H.** To solve this problem, use the fact that since $AB \cong BC$, triangle ABC is isosceles. In isosceles triangles, the angles opposite equal sides have equal measure. In this triangle, since angles A and C are opposite sides AB and BC, they have equal measure. Because the sum of the angles in a triangle is always 180°, the sum of the measures of angles A and C is 180° less the measure of angle B, or $180° - 55° = 125°$. Therefore, since angles A and C have equal measure, they both equal $\frac{125°}{2} = 62.5°$.

7. **The correct answer is D.** To solve for x, distribute the 3 in the equation $3(x-2) = -7$ to get $3x-6 = -7$. Adding 6 to both sides of the equation yields $3x = -1$, or $x = -\frac{1}{3}$.

8. **The correct answer is K.** To solve this problem, factor the polynomial $x^2+3x-18$. To do so, think of $x^2+3x-18$ as $(x+?)(x-?)$. To fill in the question marks, find two numbers that multiply to equal −18 and add up to 3. One such pair of numbers is 6 and −3. To check, make sure that $(x + 6)(x - 3) = x^2 + 3x - 18$, which it does. Of the answer choices, only $(x + 6)$ is a factor.

9. **The correct answer is A.** A line parallel to the x-axis is horizontal. Horizontal lines have the equation $y = b$, where b is a constant, and represents the y-intercept. Therefore, a line in the standard (x, y) coordinate plane is parallel to the x-axis and 5 units below it will have the equation $y = -5$.

10. **The correct answer is J.** To solve this problem, first find the common denominator, and then add the fractions. In the case of $\frac{2r}{3} + \frac{4s}{5}$, the common denominator is 3×5, or 15. To convert $\frac{2r}{3}$, multiply by $\frac{5}{5}$ to get $\frac{10r}{15}$. To convert $\frac{4s}{5}$, multiply by $\frac{3}{3}$ to get $\frac{12s}{15}$. The sum is therefore $\frac{10r}{15} + \frac{12s}{15} = \frac{(10r + 12s)}{15}$.

11. **The correct answer is C.** If a pie recipe calls for $\frac{1}{3}$ cup sugar to make one 9-inch pie, making three 9-inch pies will require $3\left(\frac{1}{3}\right)$, or 1 cup of sugar.

12. **The correct answer is G.** To solve this problem, first perform the subtractions and then evaluate the absolute values:

$$|5 - 3| - |2 - 6|$$
$$|2| - |-4|$$
$$2 - 4 = -2$$

13. The correct answer is C. Since Jeff traveled twice as far as Ryan in half the time, Jeff traveled 40 miles in 2 hours. Jeff's average speed is equivalent to 40 miles ÷ 2 hours, or 20 miles per hour.

14. The correct answer is J. To solve this problem, substitute -5 for x in the expression $2x^2 + 6x$:

$$2(-5)^2 + 6(-5)$$

$$2(25) - 30$$

$$50 - 30 = 20$$

15. The correct answer is A. To solve this problem, substitute 4 for b in the equation $b - 2 = ab + 16$:

$$4 - 2 = 4a + 16$$

$$2 - 16 = 4a + 16 - 16$$

$$\frac{(-14)}{4} = \frac{(4a)}{4}$$

$$a = -\frac{14}{4} = -3.5$$

16. The correct answer is H. Recall that the area of a parallelogram is equivalent to base × height. Since *STVX* is a square, all sides of *STVX* will be equal. Likewise, since the height of the parallelogram is equal to the height of the square, the height of the parallelogram is 3. The area of the square *STVX* is $3^2 = 9$. The area of the parallelogram is therefore $(8 + 3)(3) = (11)(3)$, or 33. Thus the ratio of the area of square *STVX* to the area of parallelogram *RUVY* is 9:33, or 3:11.

17. The correct answer is E. To solve this problem, substitute -3 for x in $2x^2 - 6x + 7$:

$$2(-3)^2 - 6(-3) + 7$$

$$2(9) - (-18) + 7$$

$$18 + 18 + 7 = 43$$

18. The correct answer is G. To solve this problem, set up a proportion between the length in inches and the length in miles represented. Let the x be the number of miles represented by 1.4 inches. Since 1.2 inches represents 50 miles, $\frac{50}{1.2} = \frac{x}{1.4}$. Multiplying both sides of the proportion by 1.4 yields $x = \frac{(50)(1.4)}{1.2} \approx 58$ miles.

19. The correct answer is A. To solve this problem, distribute the -1 and combine like terms, as follows:

$$(5x^3 + 3xz^2 - 17z) - (4xz^2 + 5z - 2x^3)$$

$$5x^3 + 3xz^2 - 17z - 4xz^2 - 5z + 2x^3$$

$$(5x^3 + 2x^3) + (3xz^2 - 4xz^2) + (-17z - 5z)$$

$$7x^3 - xz^2 - 22z$$

20. The correct answer is J. The distance on the y-axis from point A to point B is the difference in the y-coordinates of points A and B. Since point A is $(-2,8)$ and point B is $(5,3)$, the distance in the y-direction is $8 - 3 = 5$.

21. The correct answer is D. The solutions of $(x - 5)(x - 3)(x + 3)(x + 9) = 0$ are the values of x that make the product $(x - 5)(x - 3)(x + 3)(x + 9)$ equal to zero. Such values of x will be the values that make any one of the four parts of $(x - 5)(x - 3)(x + 3)(x + 9)$ equal to zero, which would be 5, 3, -3, and -9. Of the answer choices, only -5 is not a value of x such that $(x - 5)(x - 3)(x + 3)(x + 9) = 0$.

22. The correct answer is H. At first glance, you might think that you don't have enough information to solve this problem. However, if you recognize that pr must be a positive fraction since it lies between 0 and 1, you can work your way through the answer choices and eliminate those that could be true:

Answer choice F: If both p and r were less than 0, their product would be positive. It's possible for pr to be a positive fraction, because both p and r could be negative fractions, so eliminate answer choice F.

Answer choice G: If p were -1 and r were also a negative number, their product would be positive. It's possible for pr to be a positive fraction, because r could be a negative fraction, so eliminate answer choice G.

Answer choice H: If both p and r were less -1, then pr would be greater than 1, so this statement cannot be true, and answer choice H is correct.

Answer choice J: If both p and r were less than 1, their product could be positive. It's possible for pr to be a positive fraction, because both p and r could be negative fractions, so eliminate answer choice J.

Answer choice K: If p were less than 1, p could be a positive fraction. If r were greater than 0,

it would be a positive number, and it's possible for pr to be a positive fraction; eliminate answer choice K.

23. The correct answer is C. To solve this problem, recall that $10^2 = 100$. Therefore since $n = 10$, $n^2 = 100$. The number 552 can be broken up into $500 + 50 + 2$, which can be written as $5(100) + 5(10) + 2$. Again, since $n = 10$ and $n^2 = 100, 5(100) + 5(10) + 2 = 5n^2 + 5n + 2$.

24. The correct answer is G. To solve this problem, add the two equations, or multiples of them, so that cancellation occurs and you can solve for the b variable. If the equation $a + 3b = -4$ is multiplied by -3, the result is $-3a - 9b = 12$. The addition of $-3a - 9b = 12$ to the second equation is shown below:

$$3a - b = 18$$

$$-3a - 9b = 12$$

$$-10b = 30$$

By canceling out the a terms, you can solve for b:

$$b = \frac{30}{-10} = -3.$$

25. The correct answer is C. To solve this problem, first combine like terms. The expression $x + x(x + x + x)$ becomes $x + x(3x)$. Multiplying the second term yields $x + 3x^2$.

26. The correct answer is G. Since the refrigerator that sold for \$450 now sells for 7% more, the current price is \$450 + (7% of \$450), or 450+450(0.07). Remember that 0.07 is the decimal equivalent of 7%.

27. The correct answer is E. The set of all points 5 units from the x-axis in 3-dimensional space can be likened to the set of all points 5 units from the x-axis in 2-dimensional space, which are two lines. Revolving such lines about the x-axis creates a tube-like shape that is a cylinder.

28. The correct answer is J. To solve this problem, calculate the slope of the ramp in the same manner as calculating the slope of a line through the points (−2,−1) and (16,2). Slope is defined as $\frac{\text{rise}}{\text{run}}$, or $\frac{\text{change in } y}{\text{change in } x}$, therefore (−2,−1) and (16,2) the equation is:

$$\frac{(2+1)}{(16+2)} = \frac{3}{18}$$

$$\frac{3}{18} = \frac{1}{6}$$

29. The correct answer is B. To solve this problem, express the number 0.002 in scientific notation, which is 2×10^{-3}, because the decimal point moved 3 places to the right. If the number 10^{-3} is 100 times larger than some number x, then:

$$10^{-3} = 100x, \text{or } x = \frac{10^{-3}}{100}$$

$$\frac{10^{-3}}{10^2} = 10^{-3-2}$$

$$10^{-3-2} = 10^{-5}$$

$$10^{-5}$$

Therefore, 2×10^{-3} is 100 times larger than 2×10^{-5} which equals 0.00002 because you must move the decimal point 5 places to the right.

30. The correct answer is H. If a sphere has a diameter of 12, then its radius is 6. A sphere with radius 6 has volume $V = \frac{4\pi r^3}{3} = \frac{4\pi(6)^3}{3} = \frac{4\pi(216)}{3} = \frac{864\pi}{3}$, or 288π.

31. The correct answer is B. To solve this problem, first establish a common denominator and add the terms in the numerator and. For the numerator, the common denominator of $\frac{1}{3} - \frac{1}{4}$ is 12. $\frac{1}{3} = \frac{4}{12}$ and $\frac{1}{4} = \frac{3}{12}$. The result is $\frac{1}{3} - \frac{1}{4} = \frac{4}{12} - \frac{3}{12} = \frac{1}{12}$. Likewise, for the denominator, the common denominator of $\frac{1}{3} + \frac{1}{4}$ is 12. $\frac{1}{3} = \frac{4}{12}$ and $\frac{1}{4} = \frac{3}{12}$, which makes $\frac{1}{3} + \frac{1}{4} = \frac{4}{12} + \frac{3}{12} = \frac{7}{12}$. The quotient $\frac{(1/3 - 1/4)}{(1/3 + 1/4)}$ equals $\frac{(1/12)}{(7/12)}$, or $\left(\frac{1}{12}\right)\left(\frac{12}{7}\right)$, which equals $\frac{1}{7}$.

32. The correct answer is H. To solve this problem, find the least common multiple of 6 and 9. The correct answer is 18 because $6 \times 3 = 18$ and $9 \times 2 = 18$.

33. The correct answer is D. To solve this problem, first subtract 5 from both sides of $\sqrt{2x} + 5 = 9$

to get $\sqrt{2x} = 4$. Squaring both sides of $\sqrt{2x} = 4$ yields $2x = 16$, or $x = 8$.

34. The correct answer is J. To solve this problem, set up three other variables, x, y, and z. Let x be the supplement of angle p. Let y be the supplement of angle q. Let z be the angle opposite of the 105° angle, so $z = 105°$. With these new variables, you know that $p = 180 - x$ and $q = 180 - y$. This means that $p + q = (180 - x) + (180 - y)$, which simplifies to $360-(x+y)$. You can see that the three lines form a triangle and that one of the angles is 105°. That means that the other two angles, x and y must have a sum of $180 - 105$, or 75°. Thus, if $x+y = 75°$, then $p+q = 360-(x+y)$, or $360-75$, which is 285°.

35. The correct answer is E. The equation $5x - 7y = 13$ defines a line. Since there are an infinite number of points in a line, there are an infinite number of ordered pairs (x, y) of real numbers that satisfy the equation $5x - 7y = 13$.

36. The correct answer is F. To easily solve this problem, write out every possible three-digit integer: 345, 354, 435, 453, 534, and 543; six different positive three-digit numbers can be formed.

37. The correct answer is D. To solve this problem, recognize that any three angles that are part of a triangle add to 180°. Relationship I: $\angle 2$, $\angle 7$, and $\angle 12$ make up the three angles within the triangle pictured, therefore $m\angle 2 + m\angle 7 + m\angle 12 = 180°$. Relationship II: using the property that vertical angles are congruent $\angle 4 \cong \angle 2, \angle 5 \cong \angle 7$, and $\angle 10 \cong \angle 12$, therefore $m\angle 4 + m\angle 5 + m\angle 10 = 180°$. Relationship III: two of these angles in this relationship are clearly greater than 90° each, thus $m\angle 3 + m\angle 8 + m\angle 11 > 180°$. Therefore the only relationships that must be true are I and II.

38. The correct answer is G. To solve for x, substitute $y = x - 7$ into the equation $x^2 - y^2 = 49$ (Since $x - y = 7, y = x - 7$):

$$x^2 - (x - 7)^2 = 49$$

$$x^2 - (x^2 - 14x + 49) = 49$$

$$x^2 - x^2 + 14x - 49 = 49$$

$$14x - 49 = 49$$

$$14x = 98$$

$$x = 7$$

39. The correct answer is E. The expression $\frac{a^9}{a^3}$ is simplified by subtracting the exponents to get $\frac{a^9}{a^3} = a^{9-3} = a^6$.

40. The correct answer is H. To solve this problem, first determine the length of the unknown side of the polygon. To do so, use the fact that the piece cut off was a right triangle. The unknown side is the hypotenuse of the triangle with legs $14-10 = 4$ and $10 - 7 = 3$. Use the Pythagorean Theorem to calculate the length of the hypotenuse (the unknown side of the polygon):

$$4^2 + 3^2 = c^2$$

$$16 + 9 = c^2$$

$$25 = c^2$$

$$c = 5$$

The length of the unknown side is 5. Therefore the perimeter of the entire polygon is $14 + 10 + 10 + 7 + 5 = 46$.

41. The correct answer is D. To solve this problem, you must recall that a circle with center (h, k) and radius r has the equation $(x - h)^2 + (y - k)^2 = r^2$. A circle that has center $(-4, 5)$ and radius 5 units would therefore have an equation $(x + 4)^2 + (y - 5)^2 = 25$.

42. The correct answer is H. The tangent of an angle is the ratio of the side opposite the angle to the side adjacent to the angle. The side opposite angle x is 15 and the side adjacent is 8, making $\tan x = \frac{15}{8}$.

43. The correct answer is A. Enclosing a square region with 120 feet of fence would make each side of the square $\frac{120}{4}$, or 30 feet. Therefore, the number of square feet enclosed is the area of the region, which would be 30^2, or 900 square feet.

44. The correct answer is G. To solve this problem, recall that a quadratic equation has exactly one real solution when the solution is a "double zero." Such an equation would have the factored form of $(y + b)^2 = 0$, where b is a real number. When you expand the equation, it becomes $(y + b)^2$ is $y^2 + 2by + b^2$. The equation $y^2 + 2ky + 4k = 0$ has a similar form; however, in order for the term $4k$ to represent b^2, k must equal 4.

45. The correct answer is D. If the measure of angle *DCA* is 60°, then the measure of angle *BCA* is 60° as well. Because the angles in a triangle have the sum of 180°, angle *BAC* is 180 − 90 − 60, or 30°. Because the segment *AD* bisects angle *BAC*, angle *BAD* is equivalent to angle *DAC*, which is $\frac{30°}{2}$, or 15°. Further, since the triangle *ABD* is a right triangle, the measure of angle $ADB = 180 - 90 - 15$, or 75°.

46. The correct answer is F. To solve this problem, factor the numerator and denominator. Factoring the numerator $4x - x^2$ yields $x(4 - x)$. Factoring the denominator $(x^2 - 2x - 8)$ yields $(x + 2)(x-4)$. Thus the expression $\frac{(4x - x^2)}{(x^2 - 2x - 8)}$ becomes $\frac{x(4 - x)}{(x + 2)(x - 4)}$. The expression $(4 - x)$, when -1 is factored out, equals $-(x - 4)$. Therefore $\frac{x(4 - x)}{(x + 2)(x - 4)} = \frac{-x(x - 4)}{(x + 2)(x - 4)}$, which can be simplified to $\frac{-x}{x + 2}$.

47. The correct answer is C. To solve this problem, recall that circumference is $2\pi r$, where r is the radius. If a circle has a circumference of $\left(\frac{4}{3}\right)\pi$, then $\left(\frac{4}{3}\right)\pi = 2\pi r$. Therefore, $2r = \frac{4}{3}$, or $r = \frac{2}{3}$.

48. The correct answer is F. If the function f satisfies the equation $f(x+y) = f(x)+f(y)$ for every pair of real numbers x and y, then f is a linear function. For some unknown linear function f, the value of $f(1)$ can be any real number (think of all the possible lines that can be drawn).

49. The correct answer is C. To solve this problem, use the formula for the area of a trapezoid, $\frac{1}{2}h(b_1 + b_2)$, where h is the altitude (height) and b_1 and b_2 are the bases. In this case, the altitude (height) is 3 and one of the bases is 5. Because the area is known to be 24, the other base can be determined by using the equation $24 = \left(\frac{1}{2}\right)(3)(b + 5)$. Dividing both sides of the equation by $\frac{3}{2}$ (or multiplying both sides of the equation by $\frac{2}{3}$) yields $16 = b + 5$, or $b = 11$.

50. The correct answer is K. To solve this problem, it might be helpful to make a list of the powers of 2: $2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 16$, $2^5 = 32$, and so on. Select consecutive values from this list and multiply them, starting with the values that make the most sense. For example, you know that $2 \times 4 \times 8$ will be less than 512, so try $4 \times 8 \times 16$, which does equal 512. Now, add those same terms: $4 + 8 + 16 = 28$.

51. The correct answer is B. You are given that 30% of x equals 60% of y, which can be expressed mathematically as $0.3x = 0.6y$. To express y in terms of x, solve for y, as follows:

$$0.3x = 0.6y$$
$$0.3x/0.6 = y$$
$$0.5x = y$$

This is equivalent to the expression $y = 50\%$ of x.

52. The correct answer is F. To answer this question, solve the inequality for x, as follows:

$$3(x - 4) \geq 9(4 + x)$$
$$3x + 12 \geq 36 + 9x$$
$$-6x \geq 24$$
$$x \leq -4$$

Remember that the direction of the sign in an inequality switches when dividing by a negative number.

53. The correct answer is E. The first step in solving this problem is to solve $x = 6a + 3$ for a:

$$x = 6a + 3$$
$$x - 3 = 6a$$
$$\frac{x - 3}{6} = a$$

Now you can substitute $\frac{x - 3}{6}$ for a in the second equation, and solve for y:

$$y = 9 + a$$
$$y = 9 + \frac{(x - 3)}{6}$$
$$y = \frac{54}{6} + \frac{x - 3}{6}$$
$$y = \frac{54 + x - 3}{6}$$
$$y = \frac{51 + x}{6}$$

54. The correct answer is K. In a trapezoid, the bases are parallel, so $AB||DC$. You are given that $PQ||AB||DC$. By definition, the ratio of the length of DP to the length of PA is the same as the ratio of PQ to AB. In other words:

$$\frac{DP}{PA} = \frac{PQ}{AB}$$

Substitute the lengths that are given for DP, PA, and AB and solve for PQ:

$$\frac{8}{12} = \frac{PQ}{36}$$

$$36 \times \frac{8}{12} = PQ$$

$$3 \times 8 = PQ$$

$$24 = PQ$$

55. The correct answer is C. To solve this problem, substitute the given values into the given equation:

$$3365 = 225(x) - (165x + c)$$

You know that 75 units were sold, so set x equal to 75 and solve for c:

$$3365 = 225(75) - (165(75) + c)$$

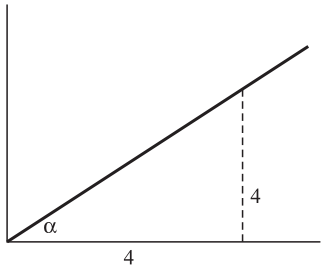
$$3365 = 16875 - (12375 + c)$$

$$3365 = 16875 - 12375 - c$$

$$3365 - 4500 = -c$$

$$1135 = c$$

56. The correct answer is J. The sine of any acute angle is calculated by dividing the length of the side opposite to the acute angle by the hypotenuse ($\sin = \frac{opp}{hyp}$). It may help you to draw a diagram to solve this problem, as shown below:



Use the Pythagorean Theorem to calculate the length of the hypotenuse.

$$a^2 + b^2 = c^2$$

$$4^2 + 4^2 = c^2$$

$$16 + 16 = c^2$$

$$32 = c^2$$

$$\sqrt{32} = c$$

$$\sqrt{16}\sqrt{2} = c$$

$$4\sqrt{2} = c$$

So, the sine of angle α is $\frac{4}{4\sqrt{2}}$, which can be reduced to $\frac{1}{\sqrt{2}}$. Recall that roots should be elimnated from denominators: $\frac{1}{\sqrt{2}}\frac{(\sqrt{2})}{(\sqrt{2})} = \frac{\sqrt{2}}{2}$.

57. The correct answer is E. Integers can be even or odd, and positive or negative. Pick real numbers to substitute into the expressions in each Roman Numeral, then eliminate any Roman Numerals that do not always yield and odd number.
Whenever a number is multiplied by 2, the result is always even. Thus, II and III are true; answer choice E is correct.

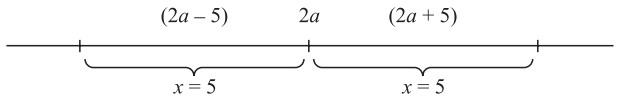
58. The correct answer is J. Probability refers to the likelihood of something happening. How many white chips should Carol put into the container so that she is likely to draw 3 red chips every eight draws? Try the answer choices.

F: If she puts in 3 white chips, then the probability of drawing a red chip is 6 (red chips) out of 9 (total chips). Eliminate answer choice F.
G: If she puts in 6 white chips, then the probability of drawing a red chip is 6 (red chips) out of 12 (total chips). Eliminate answer choice G.
H: If she puts in 8 white chips, then the probability of drawing a red chip is 6 (red chips) out of 14 (total chips). Eliminate answer choice H.
J: If she puts in 10 white chips, then the probability of drawing a red chip is 6 (red chips) out of 16 (total chips), which is equivalent to 3 out of 8, or $\frac{3}{8}$. She should add 10 white chips.

59. The correct answer is D. When a wheel makes one revolution that means that it goes completely around one time. The distance one time around a wheel is equal to the wheel's circumference.

A wheel is a circle, and the formula for the circumference of a circle is $C = 2\pi r$. If the diameter of the wheel is 27, then one time around, or one revolution, of the wheel is equal to 27, which is equal to $2r$. So, the circumference of the wheel is $2r\pi$, or, 27π. Since the wheel made 32 revolutions, multiply 32 by 27π to get 864π.

60. The correct answer is K. The absolute value of a number is indicated when a number is placed within two vertical lines. Absolute value can be defined as the numerical value of a real number without regard to its sign. This means that the absolute value of 10, $|10|$, is the same as the absolute value of -10, $|-10|$, in that they both equal 10. If the absolute value of $x-2a = 5$, then $x-2a$ must also equal -5. If $x - 2a = 5$, then $x = 2a + 5$. If $x - 2a = -5$, then $x = 2a - 5$. These equations are represented on the number line as follows:



So, the distance apart on the number line is $5 + 5$, or 10.

Reading Test Explanations

PASSAGE I

1. **The best answer is C.** Although the statement in the question uses the word "wedded," it does not simply mean "marriage;" wedded also means "closely dedicated or attached to." In this statement, Clover is implying that Cecy had an attachment or affection for pink roses, answer choice C. Answer choices A, B, and D can be eliminated because there is no reference to an actual wedding when Clover says "She *was* rather wedded to them." Clover is simply referring to Cecy's fondness for pink roses.

2. **The best answer is G.** Cecy and Clover's relationship can be implied when the passage writes that the name of Cecy's baby was "'Katherine Clover,' after her two friends." This distinguishes Clover as a close friend of Cecy, answer choice G. Answer choice F may appear to be correct; however, when the passage identifies Clover as a friend of Cecy, it is clear that they are not sisters.

3. **The best answer is D.** This statement indicates that Elsie is pleased to be sewing ruffles onto the skirt in which her sister Clover is about "to burst upon the world." The fact that Clover is preparing to "burst upon the world" tells us that Clover is attending a sort of coming-of-age or milestone event. Answer choice A simply discusses Clover's visions of her first party dress. Answer choice B is incorrect because although it discusses girls "coming out," it does not specifically refer to Clover. Answer choice C is incorrect because Elsie is dreaming of her future, which has little to do with Clover's upcoming milestone event.

4. **The best answer is G.** This statement indicates that Clover's eyes were always her most notable source of beauty; therefore, answer choice F can be immediately eliminated. The passage states that Clover's eyes, with their pathetic look, "made them quite irresistible to anyone with a tender or sympathetic heart." This indicates that her eyes were compelling to kind and compassionate people, which best correlates with answer choice G. Answer choices H and J are not supported by details found in the passage.

5. **The best answer is C.** The passage states that, "Yes, the excellent Cecy, who at thirteen had announced her intention to devote her whole life to teaching Sunday school, visiting the poor, and setting a good example to her more worldly contemporaries, had actually forgotten these fine resolutions, and before she was twenty had become the wife of Sylvester Slack, a young lawyer in a neighboring town!" This information best supports answer choice C.

6. **The best answer is H.** According to the passage, Elsie "adored Clover," which suggests that she enjoyed spending time with her sister. The context of the passage indicates that both girls enjoyed the pleasure of each other's company. The other answer choices are not supported by the passage.

7. **The best answer is B.** The passage states that "her arms and wrists and shoulders … were round and dimpled like an infant's." The other answer choices are not supported by the passage.

8. **The best answer is J.** The passage mentions several times that Cecy was trying to convince Clover to use a spray of pink roses on her dress. When Clover is discussing the type of flowers to use, she states that she wouldn't use the spray of pink roses because "it cost lots, and I didn't want it at all … fresh flowers are a great deal prettier, anyway, than the artificial ones." This clearly indicates that Clover did not want to follow her friend Cecy's advice on what type of flowers to use on her dress. The other answer choices are not supported by details found in the passage.

9. **The best answer is A.** As stated in the passage, "Elsie, who adored Clover, considered her as beautiful as girls in books." Later in the passage, Elsie is described as "the adoring Elsie." This best correlates to answer choice A, "admiring," which is a synonym for "adoring." The other answer choices are not supported by descriptions of Elsie's feelings towards Clover.

10. **The best answer is J.** In the last passage, Elsie expresses her approval of Clover's flower choice and how "lovely" she thinks that they will be. Elsie also discusses the dress that she will wear when she is old enough to "go to parties and things," and how it will be "interesting" when both Clover and herself go to pick it out. These facts both indicate that Elsie is both interested in and appreciative of Clover's opinion, answer choice J. Answer choice F is incorrect because the last paragraph highlights the fact that Clover is of age to attend parties and the like, while Elsie is not.

PASSAGE II

11. The best answer is A. In reference to Smith's *Wealth of Nations*, "the book is thoroughly studied; for the theory within, Smith's *magnum opus* remains controversial." The phrase "magnum opus" means "a great work or masterpiece," which makes sense in the context of the passage since Smith's *Wealth of Nations* is portrayed as being a far-reaching economic treatise that is studied by the masses.

12. The best answer is G. The passage states that "Adam Smith asserts the power of the 'invisible hand,' the notion that a society benefits from people acting in their own self-interest, without regard to community service." This corresponds with answer choice G. The other answer choices are not supported by details found in the passage.

13. The best answer is B. The passage discusses Smith's theory of markets and the "natural price" of goods and services, stating that "For example, should a product be in excess production, its market price would fall." If a decrease in market price is caused by having too much of a product, you can safely assume that an increase in market price would be caused by a shortage in production of a product. Therefore, answer choice B is the best choice. Answer choice A is incorrect because the passage states that stock is reduced if a product is in excess production, not during a shortage of that product. Answer choice C may appear to be correct; however, Smith's theory states that "natural prices" include a "reasonable profit" but does not indicate whether profit margins fluctuate with excesses/shortages of products.

14. The best answer is J. The passage tells you that "the butcher does not cut meat because the community desires it, but rather because it is a means to earn money … The buyer of a steak values the steak more than his money, while the butcher values the money more than the steak." Applying this to the baker, you can infer that the baker bakes in order to earn money and that the community members value the bread more than their money. This corresponds with answer choice J. All of the other answer choices contradict the example set forth in the butcher illustration.

15. The best answer is C. The passage states that Smith believed that the "natural price" of goods and services should equal "the production cost plus a reasonable profit." Since "profuse" means "extravagant or excessive," Adam Smith would not see a profuse amount of profits as being "reasonable." Therefore, answer choice C is correct. All of the other answer choices are supported by details found in the passage.

16. The best answer is J. Those people who criticize the *Wealth of Nations* "argue that untempered self-interest is immoral," or in other words, excessive self-interest is corrupt or unethical, answer choice J. The other answer choices are beyond the scope of the passage and are not illustrated as being reasons for criticisms of Adam Smith's *Wealth of Nations*.

17. The best answer is A. According to the passage, "Standing as a testament to his *benevolence*, Smith bequeathed much of his wealth to charity." "Benevolence" is a "tendency to perform kind or charitable acts," which best supports answer choice A. Answer choices B and D can be eliminated because it is implied that Smith voluntarily donated much of his wealth; the words "obligatory" and "requisite" imply that Smith was required to donate his money. Answer choice C does not make sense within the context of the passage.

18. The best answer is H. Smith's quotation begins by stating that man "intends only his own gain," which insinuates that people are naturally selfish and are out for their own interests. This eliminates answer choice J. The quotation continues on to say that man is "led by an invisible hand to promote an end which was no part of his intention." This statement tells you that people are led towards unintentionally fulfilling certain objectives by the "invisible hand." This suggests that the "invisible hand" directs the actions of selfish individuals and creates an outcome—one that ultimately benefits society—not anticipated by the individual. Answer choice G may appear to be correct, but the quotation does not indicate that the "invisible hand" discourages people from following their selfish interests; it merely says that the hand promotes some actions that are unintentional for man.

19. The best answer is C. When discussing the theory of natural price, the passage states that "This idea would become central to the doctrine of the *laissez-faire* economists several generations later." This supports answer choice C. The other answer choices are not supported by the passage.

20. The best answer is F. Answer choices G and J can be eliminated immediately; Smith was an advocate for "low government intervention" and believed that markets recover from recession "without intervention on the part of government or similar regulatory bodies." Answer choice F is

the best answer choice because in a competitive market, there are many options for comparable products. This can be paralleled to Smith's hypothesis on excessive production of products—the more available a product is, the lower its market price falls. Therefore, competition would be a good thing for consumers in keeping prices reasonably low.

PASSAGE III

21. The best answer is B. The closing sentence of the passage states that "Though rarely awarded for his many accomplishments, Nyman certainly remains an important figure in the innovation of classical music and represents a substantial reason why its popularity persists." This conveys both the sense that Nyman has produced important music and the fact that Nyman is underappreciated because he is "rarely rewarded." Answer choice A may appear to be correct; however, the passage states that "in spite of its waning limelight …" This indicates that the fading popularity of classical music is still occurring, and is not a thing of the past.

22. The best answer is H. The first paragraph states that classical music has been "pushed aside by such contemporary forms as jazz, rock, and rap." This describes the other types of music with which classical music has come to share an audience. The first paragraph also states that classical music is "always the subject of experimentation on the part of composers and performers," and the passage also mentions the "various schools of composition that emerged in the 20th century." This provides basis for the fact that classical music has not stopped evolving, and these points together provide the best support for answer choice H.

23. The best answer is C. The passage states that "Nyman's compositions vary greatly in mood and orchestration, but generally reflect the characteristic tenets of minimalist *fare*." The context of that statement suggests that Nyman's compositions typically reflect the style of the collection of minimalist music. The other answer choices are not supported by the passage.

24. The best answer is H. Answer choice H states that "These works, though, would not reach his largest audience," indicating that the works of Nyman's band were not largely popular. Nyman's popularity came later with his score for *The Piano*. Answer choice J may appear to be correct; however, answer choice J calls attention to the high point in Nyman's commercial success, not the fact that his popularity is limited.

25. The best answer is A. The passage states that Nyman "layers repetitive melodies … to create an atmosphere of soaring highs and endless sorrows." It was these emotional melodies that were said to put the audience into a trance, answer choice A. Answer choice B is incorrect because Nyman including "soaring highs," and finished the film "on an uplifting note." Likewise, the other answer choices are not supported by the passage.

26. The best answer is J. The author describes Nyman as "one of the great writers, conductors, and performers" of his type of music; the author also describes Nyman's film scores as "stunning." The author also identifies the score to *The Piano* as being one of the best-selling film musical recordings ever. Answer choices F and G can be eliminated because the author has clearly distinguished Nyman as worthy of critical acclaim, and the fact that his score to *The Piano* has been so successful proves that his music does appeal to modern audiences. Answer choice J makes the most sense within the context of the passage.

27. The best answer is D. The word "snub" means "to rebuff or refuse to acknowledge," and the author describes the score to *The Piano* in a very positive light. From this, you can infer that although the author believed that Nyman's work was very good and deserving of an Academy Award nomination, Nyman was not so well received by the Academy. The other answer choices are not supported by the context of the passage.

28. The best answer is G. The passage states that "In spite of its waning limelight, however, classical music occupies an enduring niche in Western culture, always the subject of experimentation on the part of composers and performers." This assertion highlights both the fact that classical music has never disappeared, and the fact that classical music is always changing and evolving due to the ceaseless experimenting of composers and performers. Although answer choice H may appear to be correct, this choice focuses only on the music of Michael Nyman and not classical music as a whole.

29. The best answer is C. It is stated in the passage that Michael Nyman "emerged as" one of the great originators of "experimental and often minimalist pieces of music." The passage goes on to state that it was Nyman himself "who coined the term 'Minimalism.'" This indicates that Minimalism was still very new when Nyman emerged as a composer, being that it had not even been formally named yet. However, although Nyman gave a formal name to the musical style, he did not create

this musical style. Therefore, answer choice D is not the correct answer.

30. **The best answer is G.** The passage states that "Nyman layers repetitive melodies played on string instruments to create an atmosphere of soaring highs tempered with sorrow, but these melodies overcome hopeless melancholy to finish on an uplifting note. The austere blues and greens of the film's cinematography blend with Nyman's round melodies to impose a trance on the audience, infusing a cold future reality with vivid romance." The word "austere" means "stark" or "severe." The statements suggests that the severity of the film's cinematography blended with Nyman's softer, "rounder" melodies to appeal to the audience's sense of romance. The other answer choices are not supported by the passage.

PASSAGE IV

31. **The best answer is C.** "Fascinating" and "intriguing" are synonyms of one another, both meaning "interesting or exciting." Likewise, "contentious" and "controversial" are synonyms of one another, both meaning "causing disagreement." Answer choices A, B, and D do not have pairs of words that mean the same thing as "fascinating" and "contentious."

32. **The best answer is G.** The passage states that mortar used in the construction of the Great Pyramid was found to be formed of "particles of pollen, charcoal, and other organic matter," and that most of the mortar collected was made of "processed gypsum with traces of sand and limestone." Answer choice G, processed limestone, is the only material not explicitly listed as composing the mortar of the Great Pyramid.

33. **The best answer is A.** The passages tells you that gypsum needs a temperature of 265 degrees Fahrenheit to be turned into mortar; on the other hand, "to make lime mortar, extremely high temperatures around 1800 degrees Fahrenheit are needed." This best supports answer choice A. Answer choices B, C, and D are not supported by details found in the passage.

34. **The best answer is H.** The passage states that the gypsum mortar used in constructing the Great Pyramid "was not used to bond the heavy stone blocks together." This fact supports answer choice H. The passage also goes on to state, "The gypsum mortar's role was to buffer the joints and to reduce friction as the enormous blocks were placed." This tells you that answer choices F, G, and J all were uses of the mortar when constructing the Great Pyramid.

35. **The best answer is D.** Throughout the passage, there are several references to researchers not having conclusive facts and making the best hypotheses they can regarding their findings. For example, the fourth paragraph begins with the statement "Examining the mortar from the Great Pyramid assists scientists in making inferences about Egypt's past," and goes on to discuss the conclusions researchers drew based on the findings that they had discovered. Answer choice A may appear to be correct; however, the information given on radiocarbon dating does not provide an exact date of construction. Rather, scientists had to conjecture about the precise date the Pyramid was built, which supports answer choice D. Answer choice B is incorrect because the passage states that after analysis of the gypsum mortar, "even using modern techniques, scientists have been unable to reproduce it." Answer choice C is beyond the scope of the passage; there is no evidence that the conclusions of scientific researchers and archaeologists usually vary.

36. **The best answer is J.** The passage states that Egypt "has few natural fuel sources available," and goes on to say that "The shortage of natural fuel sources would most likely have made the creation of lime mortar highly uneconomical." This best supports answer choice J. It is stated in the passage that limestone was both more durable and more abundant than gypsum, thus answer choices F and G can be eliminated. The inferences drawn by the researchers are irrelevant to the choice of material that the Egyptians used in constructing the pyramids; therefore, answer choice H is also incorrect.

37. **The best answer is A.** The passage states that "Instead of clarifying or *expunging* older theories about the Great Pyramid, the results of the study left the researchers mystified." The word "expunge" means "to get rid of or remove," so within the context of the passage, the results of the study did not clarify or eliminate any older theories about the Great Pyramid, answer choice A.

38. **The best answer is H.** The passage states that from radiocarbon dating, it was found that the average date of construction for the Great Pyramid was 2977 BC. The passage goes on to state that this discovery was disputed because this date asserted that the Pyramid was built "over 400 years earlier than most archaeologists originally believed." If archaeologists believed the Pyramid to have been

constructed 400 years after 2977 BC, the original date of construction was, therefore, assumed to be approximately 2577 BC.

39. The best answer is B. As stated in the passage, "researchers began studying the mortar from the pyramid, hoping it would reveal important clues about the pyramid's age and construction." This supports answer choice B. The topics of the other answer choices are discussed throughout the passage; however, these topics are not the chief topics the researchers were looking to learn from studying the mortar.

40. The best answer is G. The passage states that the gypsum "would have been easier to mine than limestone." This best correlates with answer choice G. Answer choice F may appear to be correct; however, the passage does not discuss whether a more water-soluble material would be advantageous when making mortar.

Science Reasoning Test Explanations

PASSAGE I

1. **The best answer is A.** If Experiment 1 were repeated with a length of 4 m, the movement time for the wood block would be expected to be greater than the time measured for 2 m (1.64 s). The only answer choice with a time greater than this is answer choice A (2.31 s).

2. **The best answer is H.** According to Table 2, the plastic object in Experiment 2 took an average of 1.41s to move down the inclined plane. This time is closest to the average time of 1.42 s in Table 3. The object rating of the object with an average time of 1.42 s is 13. Because the plastic object's time is just below the time in Table 3, it would most likely have an object rating of between 9 and 13, answer choice H.

3. **The best answer is A.** The students rated objects on a scale from 5 (least friction) to 30 (most friction). In order for the chair to move more easily across a floor, the material must have a low friction rating. Therefore, a material with an object rating of 5 (the least amount of friction) would move most easily across a floor.

4. **The best answer is J.** In Experiment 1, the volume of each of the blocks of material was 750 cm^3. In Experiment 2, the mass of each of the blocks of material was 1500 g. The difference between the two experiments is that, in Experiment 1 volume was held constant, while in Experiment 2 mass was held constant. The correct answer is answer choice J.

5. **The best answer is B.** If the loss in speed due to friction leads to longer travel time down the inclined plane, to rank objects in order of increasing loss in speed due to friction you must rank them in order of increasing time. Answer choice B is correct because glass (1.31 s) < aluminum (1.35 s) < wood (1.69 s) < concrete (2.47 s).

6. **The best answer is H.** To answer this question compare the average times listed for each block in Tables 1 and 2. When the volume is increased, the average time goes up for some blocks and down for others.

PASSAGE II

7. **The best answer is D.** Scientist 2 states that when iron is "hotter than 1043K, its Curie temperature, iron loses its magnetic properties." The scientist is implying that the Earth's core has a temperature above 1043K; therefore, it has lost its magnetic properties and cannot cause the Earth's magnetic fields. Answer choice B can be eliminated because this is the viewpoint of Scientist 1 only.

8. **The best answer is H.** Scientist 1 states: "The magnetic field grows and wanes, and the Earth's poles drift and occasionally flip as the rotation of the core changes." Scientist 2 states that the molten outer core and the Coriolis effects "generate the fluctuating magnetic field." Both scientists address the Earth's changing magnetic field and, therefore, are in agreement with the statement in the question.

9. **The best answer is C.** Both scientists talk about the fluctuations in Earth's magnetic field. They would both agree with answer choice C because it describes the varying magnetic properties in volcanic rocks as the result of the fluctuations they have talked about. Answer choices B and D can be eliminated because solar wind is only discussed by Scientist 1. Answer choice A supports Scientist 2 only.

10. **The best answer is J.** According to Scientist 1, "The inner core … consists of a solid iron ball, which would exhibit properties or *ferromagnetism* …" Scientist 2, on the other hand, suggests that the core is too hot to maintain its magnetic properties. This information best supports answer choice J.

11. **The best answer is B.** According to Scientist 1, the coordinated motions of electrons within iron atoms create the Earth's magnetic field. According to Scientist 2, the Earth's magnetic field is created from the circulating electric currents in the Earth's molten outer core.

12. **The best answer is G.** Scientist 2 says, "The Earth's magnetic field is attributed to a *dynamo effect* of circulating electric current in the molten outer core." If a new type of sensor detected electric currents in the outer core, the new technology would be consistent with the view of Scientist 2. Scientist 1, however, does not talk about the outer core.

13. **The best answer is A.** Scientist 2 says that the Earth's inner core does not influence the Earth's magnetic field because iron loses its magnetic properties at high temperatures (reached by the inner core). This contradicts what Scientist 1 says, so it is likely that Scientist 1 would consider this a flaw in Scientist 2's theory.

PASSAGE III

14. The best answer is G. According to Figure 1, 50% (half) of the Radon 222 sample is left after 4 days.

15. The best answer is C. Passage III states: "Radioactive decay is a natural process by which an atom of a radioactive isotope spontaneously decays into another element."

16. The best answer is H. According to Table 1, the energies associated with alpha particle emission were 5.590 MeV and 3.792 MeV. The energies associated with beta particle emission were 1.308 MeV and 1.533 MeV—lower than that of the alpha particle emission.

17. The best answer is B. The decay energy of 1.796 MeV is closest to the decay energies in Table 1 associated with beta particles (1.308 and 1.533 MeV). Therefore, the particle most likely emitted is a beta particle. Answer choice D can be eliminated because Table 1 provides no information about gamma rays.

18. The best answer is G. The isotopes will have the same percent of atoms remaining at the point at which the lines associated with each isotope cross. Based on Figure 1, Radon 222 and Polonium 218 cross at about Day 4.

19. The best answer is D. Because the shape of the curves in Figures 1 and 2 are unvarying parabolas, it is clear that the decay rate of these two elements is constant over time.

PASSAGE IV

20. The best answer is H. According to Table 1, the ideal lead level is below 0.015 mg/L. Sample 1 has a lead level of 0.01 mg/L, which is at the ideal lead level.

21. The best answer is C. The passage explains, "Water with a low pH (<6.5) could be acidic, soft, and corrosive." If an ideal alkalinity level prevents pH levels from becoming too low, it will also prevent water from becoming overly corrosive. Answer choice B can be eliminated because the ideal alkalinity level shown in Table 1 is between 200 and 500 mg/L.

22. The best answer is G. Table 1 shows that Sample 1's water had a pH of 6.0. Because the ideal pH level is between 6.5 and 8.5, the water in Sample 1 is too acidic, and therefore, too corrosive. Answer choice H can be eliminated because the alkalinity level of Sample 1 was below the ideal level.

23. The best answer is A. According to Table 1, Sample 2 water has the ideal levels of all of the tested factors except iron and hardness. The water tested above the ideal level for both of these factors. This water would be acceptable as long as the developers are willing to accept high iron levels and hard water. Answer choices B and C can be eliminated because only Sample 1 contained soft water.

24. The best answer is F. Sample 2 has an ideal level of alkalinity and can effectively buffer the water. Sample 1, however, has a level of alkalinity below the ideal and is susceptible to changes in pH. If the chemical additive caused a drastic increase in pH levels in un-buffered solutions, Sample 1's pH level would be unsafe but Sample 2 would be unaffected.

PASSAGE V

25. The best answer is C. The average speed on asphalt was 14.7 ft/s, which is higher than the other surfaces.

26. The best answer is J. According to Table 1, the average speed on asphalt was 14.7 ft/s, which is less than the 14.8 ft/s speed measured in Trial 3.

27. The best answer is A. To answer this question, you must solve each answer choice and see which answer is correct.

A: $14.8\left(\frac{2}{3}\right) = 9.87$ ft/s (speed in the rough)

B: $14.8\left(\frac{1}{3}\right) = 4.93$ ft/s (speed on the fairway)

C: $12.7\left(\frac{2}{3}\right) = 8.47$ ft/s (speed in the rough)

D: $9.4\left(\frac{2}{3}\right) = 6.27$ ft/s (speed on asphalt)

Therefore, the average speed of the cart in the rough is approximately two-thirds of the average speed of the cart on asphalt, answer choice A.

28. The best answer is F. In a scientific experiment or study, the independent variable is the factor that is changed by the experimenter. Each study was conducted using a different surface, so the surface upon which the golf cart was driven was the independent variable. The factor being studied or observed is the dependent variable. Whether it changes depends on the manipulation of one or more independent variables.

29. The best answer is C. The slowest travel of all of the studies and trials was recorded in Table 3, Trial 2: 55.4 seconds.

PASSAGE VI

30. The best answer is H. According to the information in Table 1, Grackle E (the heaviest bird at 350 g) was ranked 3rd—right in the middle—in aggressiveness. In addition, Table 3 indicates that Grackle A, the lightest bird, had more wins and fewer losses than did Grackle E, which would suggest that Grackle A is more aggressive than Grackle E. The data best supports the idea that body weight has no effect on aggression, answer choice H.

31. The best answer is A. Table 3 shows each of the grackles' win–loss records. The grackle with the highest win–loss record was Grackle B (95–10). This information would suggest Grackle B would have the highest likelihood of mating, answer choice A.

32. The correct answer is J. To solve this problem, compare the values given with the values listed in Table 1. Based on those values, the sixth grackle is most similar to Grackle D, which had an aggression rank of 5. However, because the sixth grackle had fewer calls, it would be less aggressive, and so would likely have a lower rank.

33. The best answer is D. According to Experiments 1 and 2, in general, the more calls a grackle makes, the higher the aggression ranking and the more times the grackle wins. This information suggests the number of calls is related to the dominance of one male grackle over other males.

34. The best answer is H. Table 2 shows the number of times each grackle won over another grackle. According to the table, Grackle A won over Grackle C a total of 5 times. Grackle C won over Grackle A 20 times. This information suggests the two birds encountered each other $20 + 5 = 25$ times.

35. The best answer is B. In order to refute the criticism that the order the grackles were placed in the cage may have affected the aggressiveness of each bird, the experiment should be repeated several times with different orders each time. Answer choice A seems plausible, but even though the order is randomized, it might still affect aggressiveness. By repeating the experiment several times, the criticism will be overcome.

PASSAGE VII

36. The best answer is G. To answer this question, you must look at Table 1 and determine the amount of growth for each week.

Plant type 1: Week 0–2: $1.2 - 0 = 1.2$ in.
Week 2–4: $2.8 - 1.2 = 1.6$ in.
Week 4–6: $3.7 - 2.8 = 1.1$ in.

Plant type 3: Week 0–2: $0.6 - 0 = 0.6$ in.
Week 2–4: $1.7 - 0.6 = 1.1$ in.
Week 4–6: $2.0 - 1.7 = 0.3$ in.

Plant type 4: Week 0–2: $0.9 - 0 = 0.9$ in.
Week 2–4: $2.6 - 0.9 = 1.5$ in.
Week 4–6: $3.5 - 2.6 = 0.9$ in.

In all three plant types, the greatest growth occurred between Weeks 2 and 4.

37. The best answer is C. In Experiment 1, Plant type 2 was not studied. According to Table 3, Plant type 2 is from the Northern Forest.

38. The best answer is H. To answer this question, find the difference in growth between Week 1 and Week 6 for each plant type:

Plant type 1: $3.7 - 1.2 = 2.5$
Plant type 3: $2.0 - 0.6 = 1.4$
Plant type 4: $3.5 - 0.9 = 2.6$

Answer choice J can be eliminated because each plant had a different amount of growth.

39. The best answer is B. To answer this question, you must look at the different in growth between the plants in Tables 1 and 2.

Plant type 1: $5.1 - 3.7 = 1.4$ in.
Plant type 3: $5.4 - 2.0 = 3.4$ in.
Plant type 4: $4.9 - 3.5 = 1.4$ in.

Fungi had the greatest affect on Plant type 3.

40. The best answer is H. According to Tables 1 and 2, Plants 1, 3, and 4 all grew more when the fungi present in the soil. This information suggests that the presence of certain fungi in the soil increases plant growth.

Writing Test Explanation

Because grading the essay is subjective, we've chosen not to include any "graded" essays here. Your best bet is to have someone you trust, such as your personal tutor, read your essays and give you an honest critique. If you plan on grading your own essays, review the grading criteria and be as honest as possible regarding the structure, development, organization, technique, and appropriateness of your writing. Focus on your weak areas and continue to practice in order to improve your writing skills.

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